

Product Requirements Document (PRD): Intercept

AI-Driven Autonomous Phone Fraud Shield

1. Document Control & Metadata

Attribute	Detail
Product Name	Intercept
Document Owner	Principal Product Manager, AI Security
Date	June 28, 2026
Version	1.0
Status	DRAFT – Pending Architecture Review
Target Release	V1.0 (Web/MVP Phase) – July 2026

2. Executive Summary & Problem Statement

The Problem: The landscape of financial fraud has fundamentally shifted from technical system breaches to psychological human exploitation. Syndicates utilize high-pressure social engineering tactics—such as "Digital Arrests," police impersonation, and tech-support scams—to bypass traditional digital banking security. Current solutions (e.g., reactive bank fraud alerts, static caller-ID blockers, and 2FA) fail because they rely on post-facto metadata or trigger *after* the victim is psychologically compromised and actively authorizing the transaction.

The Solution: Intercept is a real-time, passive AI listening shield. Rather than relying on outdated blacklists, Intercept actively evaluates the *intent* of a live conversation. By autonomously capturing and analyzing conversational audio with zero-latency speech-to-text and

intent analysis, Intercept neutralizes the threat precisely at the point of attack—blocking the call or locking down the UI before the victim can execute a wire transfer.

3. Product Vision & Goals

Core Value Proposition: To provide an invisible, zero-latency layer of defense that protects vulnerable individuals from sophisticated psychological manipulation via autonomous intervention.

High-Level Objectives:

1. **Zero-Trust Audio Analysis:** Establish a seamless pipeline capable of transcribing and scoring conversational intent in near real-time without degrading call quality.
2. **Autonomous Intervention:** Remove the burden of decision-making from the panicked user by automatically enforcing security protocols when critical risk thresholds are crossed.
3. **Privacy-First Execution:** Guarantee absolute data privacy through ephemeral, zero-retention, in-memory processing.

4. User Personas

Persona 1: The Vulnerable End-User (The Target)

- **Demographic:** Typically older adults, tech-novices, or highly compliant individuals prone to authority bias.
- **Pain Points:** Easily overwhelmed by high-pressure tactics, fearful of legal/financial repercussions, unable to distinguish between legitimate authority figures and spoofed scammers.
- **Goal:** To have an invisible safety net that requires zero active configuration or technical understanding on their part.

Persona 2: The Caregiver / Account Administrator

- **Demographic:** Adult children, legal guardians, or corporate IT administrators managing devices for others.
- **Pain Points:** Constant anxiety about their dependents falling victim to scams; discovering financial loss only after funds are unrecoverable.
- **Goal:** To deploy a reliable, autonomous shield on the dependent's device and receive asynchronous alerts if a high-risk event was intercepted.



5. Functional Requirements (FRs)

Req ID	Feature Name	Description	User Story	Priority
FR-01	Micro-Chunk Audio Capture	Frontend must capture microphone audio in continuous 3-second overlapping intervals.	<i>As an engine, I need rapid, bite-sized audio to process intent without waiting for the user to finish a long sentence.</i>	P0
FR-02	Continuous WebSocket Stream	Establish and maintain a robust <code>ws://</code> or <code>wss://</code> connection for raw byte streaming to the backend.	<i>As a system, I need an uninterrupted, low-overhead pipeline to transmit audio chunks to the processing server.</i>	P0
FR-03	In-Memory STT Pipeline	Backend must pipe raw audio bytes directly to OpenAI Whisper-1 API without writing to disk.	<i>As a privacy-conscious user, I need my voice data analyzed in memory and immediately discarded.</i>	P0
FR-04	Intent Analysis Engine	Text transcripts must be evaluated by GPT-4o using a constrained prompt for psychological pressure/scam patterns.	<i>As the system brain, I need to accurately identify urgency, manipulation, and spoofed authority.</i>	P0
FR-05	Deterministic Risk Scoring	The AI engine must return a strictly typed JSON payload containing a risk score (0-100).	<i>As a frontend client, I need a reliable, predictable data structure to trigger UI state changes.</i>	P0
FR-06	UI Intervention (Lockdown)	If Risk Score > 85%, the UI immediately transitions to a "Scam Detected" strobe/lockdown state and blocks inputs.	<i>As a targeted victim, I need the application to violently snap me out of my psychological compliance.</i>	P0
FR-07	Intervention Logs	Generate a localized, read-only log of intercepted calls (Timestamp, Threat Category).	<i>As a caregiver, I need to see a history of blocked attacks to assess ongoing threat levels.</i>	P2

6. Technical & Architecture Specifications

6.1 Architecture Workflow

1. Client Layer (Next.js):

- Leverages Web Audio API.
- UI components built with Glassmorphism aesthetic and Framer Motion for fluid, anxiety-reducing baseline states and sharp, disruptive intervention states.

2. Transport Layer (WebSockets):

- Full-duplex persistent connection. Ensures minimum TCP handshake overhead compared to standard HTTP polling.

3. Processing Backend (FastAPI):

- High-throughput asynchronous Python server.
- Manages concurrent socket connections and orchestrates 3rd-party API calls asynchronously.

4. AI Pipeline (OpenAI):

- **Whisper-1:** Handles noisy, conversational audio in various accents.
- **GPT-4o:** Executes zero-latency complex reasoning.

6.2 Deterministic JSON Schema (GPT-4o Output)

The GPT-4o engine is constrained via System Prompts and JSON-mode to output the exact following schema to ensure predictable frontend consumption:

```
{
  "risk_score": 92,
  "is_threat": true,
  "threat_category": "police_impersonation",
  "psychological_triggers_detected": ["urgency", "threat of arrest", "secrecy requested"],
  "recommended_action": "trigger_lockdown"
}
```

7. Non-Functional Requirements (NFRs)

7.1 Latency & Performance Targets

- **Total Loop Latency:** The round-trip time from audio capture completion (end of the 3s chunk) to UI intervention state change must not exceed **500ms**.
- **Processing Concurrency:** FastAPI backend must support a minimum of 10,000 concurrent WebSocket streams per node cluster without dropping packets.

7.2 Privacy & Data Governance

- **Zero-Retention Policy:** Audio data is strictly processed *in-memory* (RAM). No `.wav` or `.mp3` files are ever written to block storage or databases.
- **Compliance:** Strict adherence to GDPR and CCPA. Transcripts utilized for the intent analysis are immediately purged post-scoring. No data is utilized for model training.

7.3 Reliability & Security

- **Uptime:** 99.99% targeted availability for the core WebSocket routing and inference engine.
- **Encryption:** All data in transit must utilize WSS (WebSocket Secure) with TLS 1.3 encryption.

8. Phased Roadmap

Phase 1: MVP / Web Simulation (Current)

- **Focus:** Core AI pipeline validation, Next.js UI integration, and latency optimization.

Phase 2: Q3 2026 – Mobile App (Native Telecom Integration)

- **Engineering Prerequisites:** Deep integration with Apple `CallKit` and Android `TelecomManager` APIs.
- **Deliverable:** Background daemon application for iOS/Android that silently acts as a middle-layer during incoming cellular and VoIP calls.

Phase 3: Q4 2026 – Edge-AI Upgrade (SLM Transition)

- **Engineering Prerequisites:** Implementation of quantized Small Language Models (e.g., Llama-3 8B, Mistral-Nemo) running natively on device NPUs (Neural Processing Units).
- **Deliverable:** 100% offline detection capability. Eliminates cloud latency, bypasses internet dependency, and guarantees ultimate device-level privacy.

Phase 4: Q1 2027 – Enterprise Bank Integrations

- **Engineering Prerequisites:** ISO 20022 compliant API webhook infrastructure. B2B legal frameworks.
- **Deliverable:** Server-to-server APIs allowing Intercept to ping banking infrastructure (e.g., Chase, HDFC) to autonomously place a 24-hour hold on pending UPI or wire transfers when a >95% confidence scam is intercepted.

9. Key Performance Indicators (KPIs) & Success Metrics

Metric	Target Value	Description
False Positive Rate (FPR)	< 0.5%	The rate at which benign calls trigger a lockdown (Critical for user retention).
Detection Accuracy	> 98%	Success rate of identifying known scam methodologies within the first 60 seconds of the call.
Latency Check	< 500ms	Time elapsed between chunk transmission and UI state update.
System Uptime	99.99%	Core pipeline availability.
User Retention (Post-Intervention)	> 90%	Percentage of users who keep the app installed after experiencing an active intervention.